Haifeng Xu

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Education

Shanghai University of Finance and Economics (SUFE) **2022 – 2026**
Undergraduate of Engineering in Data Science and Big Data Technology Shanghai, China
Pilot Class of Research Institute for Interdisciplinary Science
score 89.09/100 Major score 91.62/100

Publications

Accelerated Distance-adaptive Methods for Hölder Smooth and Convex Optimization
Yijin Ren (co-first author), Haifeng Xu (co-first author), Qi Deng.
NeurIPS 2025. Link
My contributions:

- Designed the proposed algorithm.
- Completed the main mathematical proofs.

Hybrid Momentum Methods for Optimizing Deep Learning Models
Haifeng Xu (co-first author), Yijin Ren (co-first author), Chenyu Zhou, Qi Deng, Dongdong Ge, Yinyu Ye.
arXiv preprint, 2026. Link
My contributions:

- Proposed and debugged the algorithms.
- Implemented the experiments.

Research Experience

Tuning-free First Order Methods 2025/1 – Present
Mentored by Prof. Qi Deng Shanghai Jiao Tong University

- Developed accelerated distance-adaptive methods for Hölder-smooth convex optimization with anytime guarantees and no prior smoothness parameters, including a line-search-free stochastic extension.
- Established optimal leading accuracy dependence in composite convex optimization with only an additive logarithmic cost for distance adaptation, removing the multiplicative logarithm in our earlier bound.
- Research outputs:
  - Paper accepted at **NeurIPS 2025**. Link
  - Completed a separate, entirely new study; polishing the manuscript for journal submission by the end of 2026.
  - Two additional projects in progress.

Optimizers for Neural Networks 2025/9 – Present
Mentored by Prof. Qi Deng, Prof. Yinyu Ye Shanghai Jiao Tong University, Stanford University

- Demonstrated through controlled interventions that retained gradients can still reduce a target loss thousands of steps later, and that refreshing them can yield further gains.
- Found that useful gradient contributions are unevenly distributed across training examples, providing empirical support for retaining long-term gradient history.
- Proposed Hybrid Momentum (HyM) and HyMUON to exploit this history while adapting to recent gradients; improved over Muon on the evaluated translation and language-modeling benchmarks.
- Research outputs:
  - Preprint available on arXiv. Link
  - Exploring further research directions in neural network optimization.

Efficient QP Solvers Ongoing

- Currently developing an efficient quadratic programming (QP) solver based on PDHG.
- Completed the theoretical analysis and an efficient solver; further development is ongoing.

Online Robust Mechanism on Cooperative Games 2024/9 – 2025/2
Mentored by Prof. Hu Fu and Prof. Zhihao Gavin Tang SUFE

- Constructed a counterexample showing that no mechanism can achieve the approximation rate beyond $\mathcal{O}(1/k)$.
- Proved that with partial information about the value function, linear programming can be used to design a complete mechanism.

Other Experience

TA for High-dimensional Data Analysis 2025/9 – Present
Mentored by Prof. Hongsong Yuan SUFE

TA for Algorithm Design and Analysis 2025/2 – 2025/6
Mentored by Prof. Hu Fu SUFE

LLMs Technical Intern 2024/9 – 2025/1
Cardinal Operations Shanghai, China

TA for Mathematical Analysis II 2024/2 – 2024/6
Mentored by Prof. Tsz Chiu Kwok SUFE

Skills

Programming languages C++, Python, R, LaTeX

Award

The People's Scholarship In China(2023)